

Weekly Project Report –Week 6 Unified Generative AI Models via CommandLine Interface

Section-5 Group-06

Araf Tahsan Pavel ID: 2212079042

Md Tanveer Hossain Nihal ID: 2211935042

Ismot Ara Emu ID: 2212108042

Weekly Report

Objective:

Extend our unified CLI to support both vision-based generative models (DCGAN, CGAN, CVAE), audio(**WaveGAN**) and a text-generation model (GPT-2) while ensuring the codebase remains modular and conflict-free.

Last Week (Week 5):

Integrated GPT-2 Fine-tuning

- Implemented a complete GPT-2 fine-tuning pipeline using PyTorch and Hugging Face.
- Developed methods for training, text generation, and prompt handling.
- Structured scripts to efficiently manage text prompts, file outputs, and detailed result logging.

This Week (Week 6):

WaveGAN Model Development

- Designed and implemented WaveGAN Generator and Discriminator models for audio generation.
- Verified model architectures in PyTorch, tested forward propagation, and confirmed accurate tensor shapes and outputs.

Dataset Acquisition

- Downloaded and extracted the ESC-50 dataset for audio training.
- Validated the dataset directory structure and ensured audio files' availability.